

Image Classification "Threshold" — Design Discussion

Draft for the customer meeting. Goal: agree on **how to define when an image is "done"** so it leaves the classification stream and its result is reported to the researcher.

Assumption for this draft: there are **enough experts** in the system. We can therefore treat expert votes as a first-class signal (weighting, tie-breaking, even finalisation) rather than a rare edge case.

1. The problem

Today every image stays in the classification stream indefinitely — users keep being served images with no notion of "this one has enough answers." We want a **threshold**: a per-task rule that decides when an image has collected *enough* classifications. When an image crosses its threshold:

1. It is **removed from the stream** — users are no longer served it.
2. Its **aggregated result** (the agreed label) is finalised and reported to the researcher.

The open questions are: *what exactly* do we count toward the threshold, and *whose* votes count, and *how much*.

2. What we already have (no new concepts needed for the basics)

The system is already most of the way there:

Concept	Where it lives today	Notes
<code>minClassificationsPerImage</code> (default 3)	<code>Task</code> entity, per task, researcher-configurable	A <i>count</i> threshold already exists in the schema.
<code>consensusThreshold</code> (default 80%)	<code>Task</code> entity, per task	An <i>agreement</i> threshold already exists too.
Per-image / per-(image, species) vote counts	<code>ClassificationRepository</code>	We can count answers per image and per question.
<code>UserResponse</code> = YES, NO, DONT_KNOW, TRASH	<code>Classification</code> entity	Answers are NOT binary — important for the definition.
Majority vote + consensus strength	<code>CredibilityCalculator</code>	Already computes "the agreed label" and "how strong agreement is."
Per-user credibility score (0–100)	<code>User</code> / <code>CredibilityCalculator</code>	Combines gold accuracy, majority agreement, expert kappa.
Expert vs. non-expert	<code>userRole == RESEARCHER</code> marks an expert	Experts already treated specially elsewhere; repo has dedicated expert-classification queries.

The repository even has a comment marking threshold-gated completion as a known *future enhancement*. So this is mostly **wiring existing pieces together**, not building from scratch.

3. Relevant entities & parameters

Entities

- **Task** — owns the threshold configuration; researcher edits it.
- **Image** — the unit whose life-cycle ends at the threshold.
- **(Image, querySpecies) pair** — the *real* unit, because one image can be asked about multiple species. The threshold likely applies per pair, not per raw image.
- **Classification** — one user's answer (YES/NO/DONT_KNOW/TRASH) to one (image, species) question.
- **User** — has a role (expert / non-expert) and a credibility score.

Parameters the researcher might set (per task)

- `minClassifications` — how many answers are required.
- `consensusThreshold` — how strong the agreement must be to call a result (e.g. ≥80%).
- `requiredExpertVotes` — how many expert answers are required (can be 0, 1, or more).
- (Optional) which answers "count" (e.g. do DONT_KNOW / TRASH count toward the total?).
- (Optional) weighting policy — equal vote vs. credibility-weighted vs. expert-driven.

4. Open question A — *count* threshold vs. *consensus* threshold

"Is the threshold a number of YES, a number of total classifications, or an agreement level?"

Option	Definition of "done"	Pros	Cons
A1. Count of total answers	Done when image has $\geq N$ answers (any answer).	Dead simple; predictable cost per image; already in schema (<code>minClassificationsPerImage</code>).	An image with 50/50 split is "done" but has no real answer.
A2. Count of YES answers	Done when $\geq N$ users said YES.	Good if the task is "find positives."	Asymmetric — a clearly-NO image never finishes; ambiguous for multi-class answers.
A3. Consensus	Done when agreement on one label $\geq X\%$ (with a minimum number of voters).	Reports a <i>confident</i> label, not just "enough votes."	An always-split image never converges → needs a cap (see §6).
A4. Hybrid (recommended)	Done when count $\geq N$ AND consensus $\geq X\%$, OR a hard cap of M votes is reached.	Guarantees both <i>enough</i> data and a <i>confident</i> result; cap prevents images living forever.	Two/three parameters to explain to the researcher.

Note on non-binary answers: because answers are `YES/NO/DONT_KNOW/TRASH`, a pure "count of YES" definition (A2) is fragile. We should decide whether `DONT_KNOW` / `TRASH` count toward the total, or are ignored, or route the image elsewhere (e.g. flagged as junk).

Our leaning: A4 — it reuses both fields already on `Task`.

5. Open question B — whose votes count (experts, non-experts, credibility)

"Should experts and non-experts count equally? What about credibility differences?"

Because we assume enough experts, expert-driven options below are realistic, not theoretical — we can rely on experts being available for most images, not just a lucky few.

Option	How votes are weighted	Pros	Cons
B1. Equal vote	Every classification counts as 1.	Simplest; easy to explain & defend statistically.	Wastes the trust signals (expertise, credibility) we already compute.
B2. Required expert confirmation (recommended)	Crowd collects votes as usual, but the image is only "done" once it also has $\geq K$ expert votes that agree (K configurable, default 1).	With enough experts this is achievable per image; gives every finalised result a ground-truth-quality stamp; directly uses the RESEARCHER role we already track.	Slower per image; needs enough expert throughput (we assume we have it).
B3. Expert override / fast-track	A sufficient set of agreeing expert votes can finalise an image immediately , short-cutting the crowd count.	Finalises easy/clear images fast; spends crowd effort only where experts disagree or are unsure.	A small expert set can be wrong; less crowd data collected for analysis.
B4. Credibility-weighted vote	Each vote weighted by the user's credibility score (experts naturally weigh most); threshold is on the <i>weighted</i> agreement.	Smooth — no hard expert/non-expert cliff; uses the credibility engine we already built; high-trust users converge images faster.	Harder to explain; "weight collected" $\neq$ "people who saw it" → still needs a minimum distinct-humans floor.

Recommended combination: B2 + B4 — collect credibility-weighted crowd votes *and* require a small number of agreeing expert votes before finalising. The crowd gives scale and an agreement signal; the required expert vote(s) give each result an authoritative stamp; credibility weighting means a few low-trust users can't swing a result on their own.

Things to clarify with the customer:

- How many agreeing expert votes should finalise an image — 1, 2, or "experts must out-agree"?
- Should experts be able to **fast-track** (B3) clear images, or must the crowd count always be met too?
- If we weight by credibility (B4), what's the **minimum number of distinct humans** before we'll call a result, regardless of weight? (Avoids "one 99-credibility user finishes the image alone.")
- What happens if **experts disagree with the crowd majority**? (Expert wins? Keep collecting? Flag for the researcher?)

6. Cross-cutting decisions (apply to any option)

1. **Granularity** — threshold per **(image, species) pair**, not per raw image (the data is already keyed this way).
2. **Hard cap** — a maximum number of votes after which the image leaves the stream *even without consensus*, reported as "no agreement / ambiguous." Prevents immortal images.
3. **Expert routing (because we have enough experts)** — if an image needs expert votes to finish, the distribution logic should *preferentially* serve such images to experts so they don't stall waiting

for a chance expert encounter.

4. **What we report** — the finalised label (crowd majority + expert label), the agreement %, the number of voters, and how many were experts, so the researcher can judge confidence.
 5. **De-duplication** — a user must not be counted twice for the same (image, species) (already enforced via `existsByUsernameAndImage_IdAndQuerySpecies`).
 6. **Where it's enforced** — image selection already flows through one place (`getNextImageForUser` / `getNextBatchForApi`); the threshold check belongs there, so "done" images simply stop being served.
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7. Our recommendation to put in front of the customer

- **Definition (A4):** an image is done when it has $\geq N$ answers *and* $\geq X\%$ agreement *and* $\geq K$ agreeing expert votes, with a **hard cap of M** to retire stubborn images. N, X are already per-task fields; K (required expert votes) is a small addition.
- **Weighting (B2 + B4):** credibility-weighted crowd votes for scale and agreement, plus required expert confirmation for authority. Keep a **minimum distinct-humans floor** so weighting can't let one person finalise an image.
- **Expert routing:** since experts are plentiful, route expert-needing images to experts first so nothing stalls.
- **Reporting:** on crossing the threshold, finalise the (image, species) result (label + agreement %
 - voter count + expert count) and surface it to the researcher.

Questions for the customer:

1. Count-based, consensus-based, or both (our rec)?
2. Do `DONT_KNOW` / `TRASH` count toward the total, or are they ignored / flagged?
3. How many agreeing expert votes (K) should be required to finalise an image — 1, 2, or more?
4. Should experts be able to **fast-track** clearly-correct images, or must the crowd count always be met?
5. If experts and the crowd disagree, who wins — expert, keep collecting, or flag for the researcher?
6. What are sensible defaults for N (min answers), X (% agreement), and M (hard cap)?